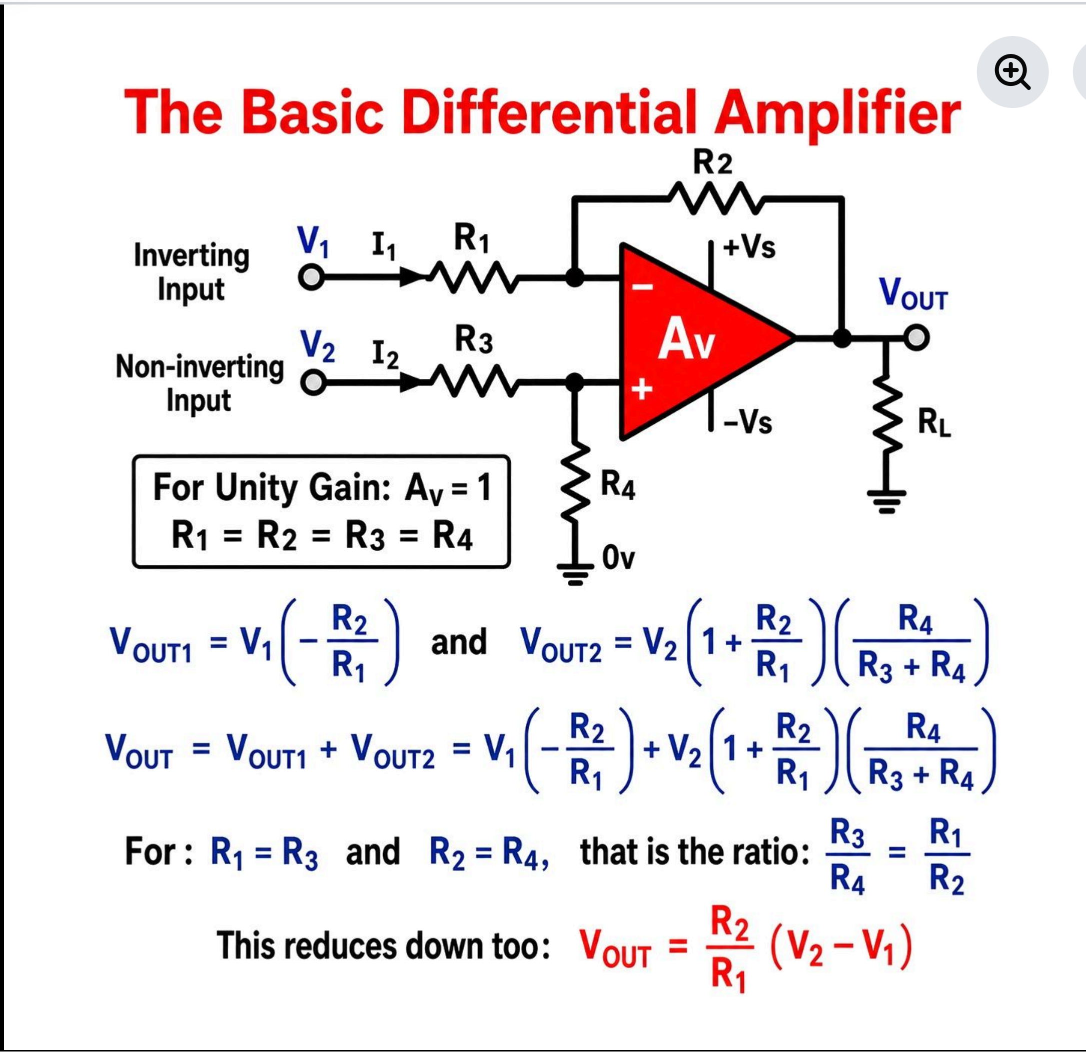

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A differential amplifier compares two input voltages and amplifies only the difference between them. In this circuit, V1 is applied to the inverting input through R1, while V2 is applied to the non-inverting input through R3. The op-amp adjusts its output so that both inputs stay at nearly the same voltage (virtual short condition).
The feedback resistor R2 controls how strongly the inverting input affects the output, while R4 forms a voltage divider with R3 for the non-inverting side. Because of this balanced structure, the circuit can reject signals that are common to both inputs, which is called common-mode rejection.
When resistor ratios are matched (R1 = R3 and R2 = R4), the circuit simplifies neatly. The output depends only on the difference between inputs: Vout = (R2/R1)(V2 − V1). This makes it very useful in sensor measurements, noise reduction, and signal conditioning where small differences need to be amplified accurately.
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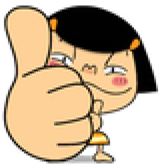

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